

What Holds a Hand-Drawn Diagram?

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Abstract

Diagram editors ship an automatic layout; MySQL Workbench calls it **Arrange > Autolayout**. Underneath is a cost, a weighted sum of complaints: boxes overlap, connected boxes sit far apart, edge lengths vary. The program moves boxes until the sum stops falling. 4,890 hand drawings were just pooled into a corpus, and the standing proposal is to learn the cost's weights from drawings like these. That only works if a finished drawing sits at a minimum of the cost. Nobody had checked at the coordinates.

The check is simple. Take a diagram a person drew and accepted. Move each box by two per cent of the drawing width, in sixteen directions, and ask whether any move lowers a criterion; a box no move improves is held. We ran this on 853 working diagrams, biological pathways and business processes, on 136 lab drawings, and, as controls, on **neato**, **dot** and ELK layouts of the same graphs.

The answer is one-sided. Edge length and stress hold almost no box a person placed, while the same test holds **neato**'s own layouts fully, so it does see a minimum where one exists. Overlap is satisfied exactly. What does hold the drawings is a criterion the costs do not carry: alignment into rows and columns. It holds a fifth to most of the median hand diagram and almost nothing in any optimised, random or grid-snapped layout. A hand layout is a layered layout with partial alignment, and the 4,890-drawing corpus shows the same profile.

A cost fitted to hand coordinates therefore learns the trade-offs between its own terms, not the drawer's. Whether the person aligned the boxes or the editor's guide did, the coordinates cannot tell; for fitting it makes no difference, the term is held and missing either way. The fix is an addition: offer alignment among the terms, keep length and stress, whose preference evidence stands, and report how many boxes the fitted cost holds at layouts people accepted.

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Related Version A full version with every sensitivity inline is maintained beside this one.

Supplementary Material Code, parsed corpora and every table's script: <https://github.com/Anode1/cjitter>, directory `example/diagrams`.

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1 Introduction

A layout program places the boxes of a diagram by minimising an *energy*, a weighted sum of quantities the field has agreed are bad: edges that cross, boxes that overlap, edges of very unequal length, boxes that sit on edges they do not belong to [3, 2, 4, 5]. The weights are asserted by the tool's author and reach the user as one unparameterised command: MySQL Workbench documents **Arrange > Autolayout** in full as "Automatically arranges objects on the canvas", and Oracle SQL Developer Data Modeler's documented remedy for a bad result is Undo. Since 1994 the proposal has been to learn the weights instead, from layouts people made or preferred [15, 26, 14, 19, 28]. Such methods differ in what they need of their examples. Learning from preferences or selections [15, 26, 14, 28] needs only that an accepted



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XX:2 What Holds a Hand-Drawn Diagram?

layout rank well under the learned energy. Inverse optimisation, which fits an objective to solutions taken as optimal [22, 23, 24], needs the stronger property: that the coordinates themselves sit at a minimum. The stronger property has never been tested against human coordinates, no published layout method yet commits to it, and 4,890 hand-made drawings were just collected into one corpus [12], so the first coordinate fit is one download away. That property separates an energy describing where a person put a box from one describing how they ranked the result. Where it fails, the fitted weights are the criteria’s mutual trade-offs rather than the person’s, and the term the person did optimise stays unlearned because it was never offered.

The assumption has a direct test, one criterion at a time. Take a diagram a person drew. Move one box a small distance in each of several directions and ask whether any of the moves lowers the criterion. If none does, the box is *held*: the person has put it where the criterion wants it, or where the criterion has stopped caring. Do this for every box and the fraction held says how much of the layout is a local minimum of that criterion; the criterion’s value says how well it is met. A criterion that holds every box at value zero is one the person satisfied; one that holds no box is one the person did not consider, or traded away.

The test puts coordinates and judgement on opposite sides. People prefer low-stress drawings and read them better [11, 17], while the drawings people make carry 3 to 9 times the force-directed stress in those studies’ own tables [17, 16]; neither side ran a test at the coordinates. Here the coordinates people drew are held by no distance term at all in any corpus, while *neato*’s stress layouts are held at 1.00 under the same test. People prefer what they do not draw, with the caveat that preference and production are measured on different graphs and tasks; no study yet takes both from one corpus.

We ran the test on 853 diagrams of 15 to 40 boxes from three unrelated communities (Section 4), and on layouts of the same graphs by four tools. Section 6 carries the numbers; the findings are these.

1. Overlap holds every box of the median hand diagram at value zero in every corpus, as it does in every tool that removes overlaps; the term is nonzero in 4 to 8% of hand diagrams. People satisfy it exactly.
2. Length and stress hold a hand-placed box almost nowhere, median diagram 0.00 in all three corpora and 25 to 78 boxes of roughly 7,000 each, while *neato*’s stress layouts are held at 1.00. No reweighting repairs it: a weight of 0.001 on either term drops the held fraction of the sum below 0.015.
3. Alignment into rows and columns holds 0.52, 0.21 and 0.87 against *neato*’s 0.06 and 0.07 for a grid-snapped random layout. The base energy omits it and the hand layouts carry it, by the person’s hand or the editor’s guide (Section 7), and BPMN’s lanes cannot impose it.
4. Reading direction does not close the gap. Flow holds 0.59 to 0.85 by hand where *dot* and *ELK* hold 1.00: the person tolerates a backward edge the layered tools remove.
5. Crossings and node–edge separation hold every box because they are flat at this radius, not because they are met.
6. Orthogonality is partly held: 0.19, 0.17 and 0.73 by hand.
7. The corpus of the standing proposal behaves the same: on its 1,072 drawings in the band, length and stress hold 0.00 and alignment 0.87.

The rest of the paper is the foundation behind the test (Section 2), the instrument (Section 3), the corpora, the method, the measurements, the alternatives not yet excluded (Section 7), and what is already known (Section 8).

2 Foundations

2.0.0.1 The energy.

A layout of a graph with N nodes is a point $x \in \mathbb{R}^{2N}$, two coordinates per node, the boxes' sizes held fixed. An aesthetic energy is

$$E_w(x) = \sum_{k=1}^K w_k T_k(x), \quad w_k \geq 0, \quad (1)$$

where each term T_k is a quantity to be made small. The tool's author asserts the w_k ; the field of multicriteria layout is the choice of the T_k and the w_k [2, 4].

2.0.0.2 Local minimum.

A point x is a local minimum of E if $E(y) \geq E(x)$ for every y within some distance r of x ; for differentiable E this implies $\nabla E(x) = 0$. Two terms here are not differentiable, crossings being a count and overlap having corners where boxes touch, so the test uses no gradient. Fix a radius d and a set V of unit directions in the plane (sixteen, equally spaced). For node i and direction v let $x + d v^{(i)}$ be the layout with node i moved by d along v and every other node fixed. Node i is *held* at radius d if

$$E(x + d v^{(i)}) \geq E(x) \quad \text{for every } v \in V, \quad (2)$$

and the estimand is

$$q_d(x) = \frac{1}{N} \# \{ i : \text{node } i \text{ is held at radius } d \}. \quad (3)$$

$q_d = 1$ says the layout is a local minimum of E over single-node moves of length d ; for smooth E and small d it says $\nabla E(x) = 0$ to within the resolution of V . Equation (2) compares two energies, so it is defined for every term and, evaluated per term, separates the criteria a person satisfied from those they did not. A capped descent, this study's first estimand, answers the same question for the sum only; it is retained as a secondary.

2.0.0.3 The inverse problem.

Given layouts $x^{(1)}, \dots, x^{(G)}$ that people accepted, inverse optimisation asks for the weights w under which they are nearest to stationary [23, 24]. With the directional differences $D_{iv} T_k = T_k(x + d v^{(i)}) - T_k(x)$ in hand, node i is held under w exactly when $\sum_k w_k D_{iv} T_k \geq 0$ for all v , a set of linear inequalities in w . The residual is the smallest total violation over w on the simplex, a linear program, and it is zero only if some weighting of the terms makes the layout a local minimum. A term with $D_{iv} T_k < 0$ for some v at almost every node (uniform edge length, at any layout whose edges are not all the same length) cannot take positive weight without violating the inequalities at those nodes, which is the zero weight the random-search fit found.

2.0.0.4 Step terms.

A count or a step function holds a node wherever it is flat, so its value is reported beside q_d (Appendix C).

XX:4 What Holds a Hand-Drawn Diagram?

115 2.0.0.5 The reference length.

116 The length and stress terms compare distances with a reference L . L fixed at the layout's
117 median edge length is in general not the scale at which the term is least for that layout,
118 so at a true minimum every node still has a scaling direction that lowers the term and the
119 converged reference is not held (neato's stress layouts: $q_{0.02} = 0.07$ to 0.12). L is therefore
120 the value at which the term is least for the layout as loaded, $\sum \ell^2 / \sum \ell$ over edges for length
121 and the same over $\|x_i - x_j\| / \delta_{ij}$ for stress, fixed before any move; neato's stress layouts are
122 then held at 1.00 in every corpus. The median and $1/\sqrt{N}$ are sensitivities.

123 The fit is Smelser, Miller and Kobourov's scale-normalised stress [9], Ahmed et al.'s for
124 the weighted form [10]; the pathology it repairs is theirs too (Appendix D).

125 2.0.0.6 The null and the controls.

126 A test that can only succeed is not a test. The *stationarity reference*: q_d on a layout a descent
127 has converged on must be near one, or the test lacks power for that energy; neato's layouts
128 are the reference for stress (1.00 in every corpus) and the uncapped climber's end point for
129 the sums (median 1.00 for every energy and corpus, Section 6). The *tool controls* of Section 5
130 say what a layout that does minimise something scores on each term. The *matched-budget*
131 *control*: uniform random placement at the same number of energy evaluations, which the
132 hand layout must beat, or "held under overlap" would be a property of sparse graphs and
133 not of the person.

134 3 The instrument

135 The descents, the control and the paired statistics come from `cjitter`, a C library of
136 four stochastic searches over a box of real variables: hill climbing with restarts, simulated
137 annealing, a genetic algorithm, and uniform sampling as the control. Every run is byte-
138 reproducible from its seed on every platform, every method constant is a visible field, and
139 `cjitter_compare` runs all four at the same budget on the same seeds and reports an exact
140 paired sign test, Holm-corrected, against the control. "Better than uniform sampling at equal
141 cost" is the only claim it lets a search make.

142 Its `block` operator, one object moved a random step and the step kept when the summed
143 energy fell, is the operator of a label placer the author deployed in 2001; the directional test
144 of equation (2) is that proposal made exhaustive, every direction tried, none accepted, the
145 answer read off.

146 The library was written for a diagram task. MySQL Workbench's reverse-engineering
147 rebuilds a schema diagram auto-placed after every database change, and restoring a 44-table
148 diagram by hand cost the author about an hour each time, so `cjitter` places a migration's
149 new tables onto the frozen rest. Pinning the rest is justified only if the person's positions
150 carry something worth keeping; what they carry, at the coordinates, is this paper's question.

151 4 The corpora

152 Three collections of diagrams whose node coordinates a person chose, from three communities
153 with different authoring tools, all redistributable.

corpus	licence	files	parsed	15–40 nodes	used	median m/n
WikiPathways GPML, <i>Homo sapiens</i>	CC0	1,016	571	305	305	1.06
Reactome SBGN-ML	CC0	1,382	710	248	248	1.04
BPMN Academic Initiative	academic	29,810	21,784	6,723	300	1.11

The layouts are working documents: Reactome diagrams are laid out by curators [32], WikiPathways pathways are drawn in PathVisio [33], the BPMN models are students' work in Signavio [34, 35]; nobody was asked to draw nicely, and the three tools share no layout code. Two WikiPathways files (WP5037, WP3391) carry coordinates converted from Reactome [36] and are dropped.

The BPMN archive is mixed: of the 21,784 models the parser reads, 13,435 have every edge a BPMN flow or association and the rest are EPC, Petri net and UML models, whose flow elements carry stencils outside the connector set and so become degree-2 nodes under the node rules. Only the 13,435 enter, 6,723 of them in the band, and the first 300 by model id are used; the two biological corpora are used whole.

A diagram enters as its largest connected component; container glyphs (SBGN compartments, BPMN pools and lanes) are not nodes; SBGN process glyphs are nodes, so the Reactome graphs are bipartite and their m/n is the process graph's; each layout is rescaled so that its bounding box fits the unit square, uniformly by the larger extent, so aspect is preserved, box sizes scaled with it. The directional test runs on 15 to 40 nodes; the climber pilot used 10 to 400. The registered deduplication pass finds one Reactome drawing stored under two ids and two structural near-pairs, none moving a median; clustering the WikiPathways interval by its 153 first authors widens it from [0.48, 0.54] to [0.48, 0.55] (Appendix J). The 41 to 100 node band is run in Section 6.

How the coordinates were made matters for the alignment criterion. Of the raw coordinates, 17% are integers in WikiPathways, 67% in Reactome and 63% in BPMN (44% multiples of five); the fraction of boxes sharing an exact x with another box is 0.37, 0.06 and 0.42, an exact y 0.29, 0.12 and 0.76, and either 0.54, 0.17 and 0.88, which is the figure the alignment row of Section 6 should be read against. In WikiPathways exact alignment is twice as common as integer coordinates, so it is not a rounding grid's act; the coordinates cannot say whether it is the person's hand or the editor's snap-to-object guide, which PathVisio draws on every drag. BPMN authors work on a grid inside lanes, hence the y figure of 0.76, and their editors guide every drag toward the neighbours; Reactome's curation tool rounds and its authors align little. Section 7 takes this up.

5 The method

5.0.0.1 Terms.

With n nodes, m edges, box i of width w_i and height h_i , L the reference length of Section 2, fitted to the layout under test:

The formulas, one line per term, are Appendix A; the comments in `energy.h` are normative.

The base energy is $T_C + T_O + T_L$ with weights (1, 1, 1), the form of the simulated-annealing layout of Davidson and Harel [2] and of the multicriteria tradition since [29]. The field's other default is stress [30, 31], $\sum_{i < j} (\|x_i - x_j\| - L \delta_{ij})^2 / \delta_{ij}^2$ with δ_{ij} the graph distance, which force-directed tools and `neato` minimise and which Dwyer et al. [17] found human choices track; it is the second base energy. T_R , T_A and T_N are the candidates for what the base energy misses. The alignment definition above is HOLA's gridiness [13], the published form, a step function of position. Three continuous forms are the sensitivities: A1, per node the

XX:6 What Holds a Hand-Drawn Diagram?

200 distance to the nearest other node along whichever axis is nearer, which has a corner at exact
201 alignment; A2, rows and columns priced separately; A3, a Gaussian kernel of width $s = 0.02$,
202 smooth. The corner holds the box: A1 holds 0.52 of WikiPathways boxes and A3 holds 0.20.

203 5.0.0.2 Routes.

204 An edge is the polyline the file stores: the attachment point on the source box, the interior
205 waypoints, the attachment point on the target box; where the file stores nothing it is the
206 chord between the centres. Crossings, orthogonality and node-edge separation are evaluated
207 on the route; length and stress on the centres. When a node moves its attachment points
208 move with it and the waypoints stay. The stored-route statistics are Appendix D; the chords
209 are the sensitivity.

210 5.0.0.3 Reading direction.

211 T_F needs a direction u . It is one of the four axis directions, the one under which T_F is least
212 for the layout as loaded, fixed before any move, the same rule as for L . Undirected edge
213 kinds (GPML graphical lines, BPMN undirected associations) do not enter it.

214 5.0.0.4 Directional test.

215 Radius $d \in \{0.005, 0.01, 0.02, 0.05\}$ of the drawing width, sixteen directions, every node, each
216 term alone and the sums of the tables. Primary cell $d = 0.02$. The directions are exact to
217 double precision (a series evaluated in the first quadrant and rotated), and a move that lowers
218 the energy by less than 10^{-12} of its value is a tie; an earlier build with directions off by up
219 to 4×10^{-6} counted the shift that error made as an improvement and read alignment lower
220 by up to 0.04. Deterministic: no seed, no budget. Every median carries a 95% bootstrap
221 interval over diagrams (1000 resamples).

222 5.0.0.5 Tool controls.

223 The same graphs with the same box sizes and edge directions laid out by `neato` (`-Gstart=1`,
224 so the file reproduces), by `neato` with `-Goverlap=false` (prism), by `dot`, and by ELK
225 layered 0.12.0 (`elkjs`, direction RIGHT, every other option at its default), the engine the
226 BPMN and KIELER editors run; a check in the binary confirms each control is the same
227 graphs.

228 5.0.0.6 Descent, secondary.

229 Through the library: hill climbing from the hand layout, one node per proposal, each node
230 kept within a disc of radius d about its start, 4000 energy evaluations, 15 seeds; uniform
231 sampling in the same cap at the same budget on the same seeds, the matched-budget control;
232 the converged reference the same climber uncapped, 60,000 evaluations, q_d measured where
233 it stops. The step schedule, the reported columns and the pilot's step lesson are Appendix E.

234 5.0.0.7 Fitting.

235 Weights on the simplex, the linear program of Section 2 over every node and direction
236 of a corpus (about 120,000 inequalities, HiGHS), the residual its value per node; because
237 the residual depends on the terms' scales and q does not, q under the fitted weights is
238 reported beside it, on the fitting diagrams and held out under 5×2 cross-validation (five

239 seeded halvings, each half fitted and scored on the other). Inverse optimisation is degenerate
240 whenever a term is satisfied exactly: overlap holds every node, so any weight vector with all
241 its weight on overlap has residual zero and the program returns it. The informative output is
242 the sweep, q as the weight on one term grows from 0 with the rest on crossings and overlap.

243 5.0.0.8 Statistics.

244 The unit of inference is the diagram. Per corpus, hand against a control is the Wilcoxon
245 signed-rank test with the exact sign test beside it and the Hodges–Lehmann shift with
246 its distribution-free 95% interval, Holm-corrected within the (corpus, control) family; the
247 mechanics and the script’s 630 self-checks are Appendix F. Equivalence claims (“no reweighting
248 repairs it”) are read off the sweep, never off a failure to reject.

249 6 Measurements

250 The tables in this section are from one deterministic battery on the corpora of Section 4
251 (`results/run_all.sh`; commit, versions, seeds and corpus hashes in `results/manifest.txt`);
252 the population, null and HOLA numbers ship with their scripts beside their committed
253 outputs. The directional test is one deterministic pass over the whole band on the stored
254 routes with the fitted L .

255 6.0.0.1 Each criterion alone.

256 $q_{0.02}$ on the hand layout, with its 95% bootstrap interval over diagrams, and on the three
257 tool layouts of the same graphs; in parentheses the median value of the term at the layout, 0
258 when satisfied.

XX:8 What Holds a Hand-Drawn Diagram?

corpus	energy			hand	neato	prism	dot	elk
WikiPathways	crossings alone	1.00	[1.00, 1.00]	(0.000)	1.00 (0.000)	1.00 (0.000)	1.00 (0.029)	1.00 (0.036)
	overlap alone	1.00	[1.00, 1.00]	(0.000)	0.60 (0.030)	1.00 (0.000)	1.00 (0.000)	1.00 (0.000)
	length alone	0.00	[0.00, 0.00]	(0.219)	0.24 (0.011)	0.19 (0.014)	0.00 (0.241)	0.00 (0.127)
	stress alone	0.00	[0.00, 0.00]	(0.217)	1.00 (0.042)	0.94 (0.044)	0.00 (0.201)	0.00 (0.241)
	orthogonality alone	0.19	[0.18, 0.21]	(0.157)	0.00 (0.280)	0.00 (0.281)	0.19 (0.244)	0.19 (0.158)
	alignment A1 alone	0.52	[0.48, 0.54]	(0.003)	0.06 (0.011)	0.07 (0.010)	1.00 (0.000)	0.57 (0.001)
	alignment A3 alone	0.20	[0.18, 0.23]	(0.284)	0.03 (0.446)	0.03 (0.451)	0.45 (0.204)	0.35 (0.176)
	gridiness alone	0.87	[0.84, 0.89]	(0.424)	0.78 (1.000)	0.79 (1.000)	1.00 (0.125)	0.76 (0.200)
	node-edge alone	0.88	[0.86, 0.90]	(0.004)	0.89 (0.000)	1.00 (0.000)	0.78 (0.011)	0.74 (0.016)
flow alone	0.59	[0.56, 0.62]	(0.131)	0.50 (0.211)	0.50 (0.211)	1.00 (0.000)	1.00 (0.000)	
Reactome	crossings alone	1.00	[1.00, 1.00]	(0.000)	1.00 (0.000)	1.00 (0.000)	1.00 (0.056)	1.00 (0.066)
	overlap alone	1.00	[1.00, 1.00]	(0.000)	0.76 (0.022)	1.00 (0.000)	1.00 (0.000)	1.00 (0.000)
	length alone	0.00	[0.00, 0.00]	(0.234)	0.35 (0.005)	0.26 (0.009)	0.00 (0.232)	0.00 (0.205)
	stress alone	0.00	[0.00, 0.00]	(0.212)	1.00 (0.034)	0.85 (0.042)	0.00 (0.177)	0.00 (0.197)
	orthogonality alone	0.17	[0.15, 0.18]	(0.193)	0.00 (0.277)	0.00 (0.277)	0.15 (0.256)	0.12 (0.203)
	alignment A1 alone	0.21	[0.19, 0.24]	(0.004)	0.06 (0.009)	0.06 (0.010)	1.00 (0.000)	0.42 (0.002)
	alignment A3 alone	0.10	[0.08, 0.12]	(0.300)	0.03 (0.430)	0.03 (0.428)	0.39 (0.206)	0.31 (0.213)
	gridiness alone	0.65	[0.62, 0.68]	(0.640)	0.77 (1.000)	0.75 (1.000)	0.95 (0.176)	0.74 (0.306)
	node-edge alone	1.00	[0.94, 1.00]	(0.000)	0.93 (0.000)	1.00 (0.000)	0.80 (0.007)	0.75 (0.021)
flow alone	0.62	[0.60, 0.67]	(0.096)	0.45 (0.217)	0.44 (0.222)	1.00 (0.000)	1.00 (0.000)	
BPMN	crossings alone	1.00	[1.00, 1.00]	(0.000)	1.00 (0.000)	1.00 (0.000)	1.00 (0.000)	1.00 (0.000)
	overlap alone	1.00	[1.00, 1.00]	(0.000)	0.41 (0.073)	1.00 (0.000)	1.00 (0.000)	1.00 (0.000)
	length alone	0.00	[0.00, 0.00]	(0.291)	0.42 (0.006)	0.27 (0.014)	0.00 (0.250)	0.00 (0.273)
	stress alone	0.00	[0.00, 0.00]	(0.203)	1.00 (0.017)	0.90 (0.024)	0.00 (0.169)	0.00 (0.190)
	orthogonality alone	0.73	[0.70, 0.75]	(0.011)	0.00 (0.280)	0.00 (0.278)	0.26 (0.167)	0.25 (0.139)
	alignment A1 alone	0.91	[0.87, 0.97]	(0.000)	0.06 (0.011)	0.06 (0.011)	0.89 (0.000)	0.62 (0.001)
	alignment A3 alone	0.31	[0.29, 0.33]	(0.207)	0.00 (0.453)	0.00 (0.452)	0.19 (0.168)	0.20 (0.155)
	gridiness alone	0.95	[0.94, 0.97]	(0.176)	0.78 (1.000)	0.78 (1.000)	0.79 (0.219)	0.69 (0.250)
	node-edge alone	1.00	[1.00, 1.00]	(0.000)	1.00 (0.000)	1.00 (0.000)	0.82 (0.003)	0.76 (0.016)
flow alone	0.85	[0.82, 0.87]	(0.045)	0.62 (0.121)	0.60 (0.123)	1.00 (0.000)	1.00 (0.000)	

290 6.0.0.2 The sums.

291 The same test under the summed energies, equal weights; the grid is Appendix B. Read
 292 down a hand column. Overlap holds every box at value zero; so it does in every tool layout
 293 that removes overlaps, and **neato** without removal is the only layout it does not hold, so
 294 crossings and overlap together separate a hand layout from that configuration and from
 295 nothing else. Length and stress hold almost no hand-placed box, 0.3 to 1.1 per cent per
 296 corpus, and **neato**'s stress layouts are held under stress at 1.00, so the zero is the layout's
 297 and not the test's; **dot**, which is layered and prices no distance, scores 0.00 on both as the
 298 hand layouts do. Hand and **neato** part furthest on alignment A1, and **dot** and **ELK** score
 299 high on it too because a layer is a row; a layout snapped to the same grid with no author
 300 holds 0.065, 0.075 and 0.067 (Section 7). Under T_A , q counts the fraction of boxes within
 301 $d/2$ of sharing a coordinate, not the fraction exactly on one; the term has its corner there, so
 302 it tracks the exact-tie fractions of Section 4 to within a few points.

303 Crossings, overlap and node-edge are flat at this radius (their decrease and active columns:
 304 Appendix C), and their q is not read beside The three terms above the rule are the flat
 305 ones and their q should not be read beside the rest: on BPMN only 3% of diagrams offer a
 306 crossing-reducing move and 7% an the corpus's near-trees, not about the author. The five
 307 other terms are live everywhere, and the corpus's near-trees, not about the author. The five
 308 below are live everywhere, and there the 0.00 of length and stress is a real reading: every
 309 diagram has improving moves, the mean best one worth 1.2 to 1.3 per cent of the length
 310 term and 0.5 to 0.7 per cent of stress, and none of them was taken. One such move is below
 311 perception, so at any perceptual ϵ the layouts are ϵ -stationary under the distance terms; the
 312 failure is cumulative, not local, since from the same starts the capped climber removes 0.26
 313 to 0.31 of $C+O+L$. A coordinate fit needs exact stationarity, and the layouts sit a quarter of

the energy from it. On a displacement scale, a hand layout is as far from a stress minimum as a converged layout whose every box was moved by about the test radius (Appendix E).

The BPMN figure is the one number in the paper that the corpus's selection moves. The 300 models are the first by id of the 6,723 in the band, and on all 6,723 the alignment hold is 0.87 rather than 0.91, the mean 0.81; resampling puts the committed corpus at the 99th percentile of a random 300's [0.83, 0.90]. Every other term is where it was: crossings, overlap, node-edge and $C+O$ at 1.00, length, stress and $C+O+L$ at 0.00, orthogonality 0.733 against 0.726, A3 0.312 against 0.310 and flow 0.842 against 0.854 (`results/bpmn_population.md`). The tables here stay on the committed 300; a seeded random 300 with all four tool controls regenerated (`data/bpmnr.txt`, `results/bpmn_random300.md`) confirms the confinement: A1 0.846 around the population 0.872, everything else within 0.017. One pair changes verdict: on the random 300 hand reads -0.05 $[-0.10, -0.00]$ against `dot` on A1 (Holm $p = 0.042$), below `dot` as in the biological corpora; the committed sample's apparent parity was the selection's. The alignment claim is stated at 0.87, the population value. On the value the alignment comparison survives the step terms' flatness: 58, 36 and 82 per cent of hand boxes sit in an alignment of three or more, none of `neato`'s; the value-level reading of gridiness, node-edge, crossings and flow is Appendix C.

6.0.0.3 Sensitivities.

The radius, the direction count, the reference length, the box extent measured border to border, the tie tolerance and the routes closed to chords are each swept in Appendix D; the 1,008-cell grid over them flips no verdict. The A1 levels are sixteen-direction upper bounds; the gap to every control survives at 64.

6.0.0.4 Descent and chance.

The secondary estimand's table is Appendix E; its load-bearing line is the power: the converged reference's median q is 1.00 for every energy and corpus, and the capped climber beats matched-budget uniform sampling on all 15 seeds, surviving Holm.

6.0.0.5 Reweighting.

The bracket beside each held-out q here is the range of the ten 5×2 values, a spread over correlated splits, not a confidence interval. The linear program is degenerate as Section 2 says it must be: on the two biological corpora it puts all weight on overlap (residual per node 0.0001 and 0.0000; held-out q 0.987 [0.982, 0.993] and 0.994 [0.990, 0.998]), on BPMN all weight on crossings (0.0009; 0.992 [0.979, 0.998]). Confined to length and stress it puts everything on stress and holds 0.003 to 0.011 of nodes: no distance-only weighting holds a hand layout. Confined to the four terms no layout satisfies exactly (orthogonality, alignment, node-edge, flow) it is identified, and what it identifies is the person's term: alignment carries 0.95, 0.92 and 0.89 of the weight with the held-out q stable at 0.248 [0.236, 0.260], 0.149 [0.142, 0.157] and 0.660 [0.647, 0.674]. Offered all eight terms it returns to the degenerate vertex where one is open (WikiPathways' held-out q swings [0.235, 0.991]); on BPMN, where no term is exact, it places 0.86 on alignment, stable at 0.514. An earlier draft ran this fit on the smooth A3 kernel and concluded that nothing is identified; that conclusion was the kernel's, not the data's. The sweep: the fraction of the corpus's nodes held as a weight w on one term joins crossings and overlap at equal weight:

XX:10 What Holds a Hand-Drawn Diagram?

		WikiPathways		Reactome		BPMN	
	term added to C+O	$w=0.001$	$w=1$	$w=0.001$	$w=1$	$w=0.001$	$w=1$
356	none ($w = 0$)	0.94		0.97		0.98	
357	length	0.008	0.004	0.010	0.007	0.012	0.009
358	stress	0.005	0.003	0.008	0.006	0.014	0.011
359	alignment A1	0.506	0.516	0.243	0.238	0.811	0.821
360	flow	0.576	0.606	0.613	0.625	0.797	0.810

363 The fall is a step, not a slope: whatever weight length or stress carries, 0.001 or 1, the
364 sum holds under 0.015 of the nodes, because at almost every node some direction lowers the
365 distance term and nothing else moves. Alignment and flow can carry weight: the criteria
366 people partly satisfy survive in the sum at the fraction they are satisfied. Per diagram the
367 same rows carry intervals: at $w = 0.001$ the median held fraction is 0.00 [0.00, 0.00] under
368 either distance term and 0.52 [0.47, 0.54], 0.21 [0.19, 0.25] and 0.89 [0.86, 0.95] under A1
369 (`results/constrained.py`). Under the smooth A3 kernel the alignment row reads 0.29,
370 0.13 and 0.38 at $w = 0.001$: a differentiable proxy carries a third to half of what the corner
371 carries, the number a gradient-trained consumer of this recommendation should expect.

372 6.0.0.6 Lanes.

373 82% of BPMN boxes sit in a lane or pool, and a horizontal lane constrains y , so the A1
374 hold of 0.91 could be the container's (`results/lanes.py`). Per held box, the alignment that
375 holds it: 30% a column, which no horizontal lane imposes; 13% a row shared with a box of
376 another lane or of no lane, which no lane imposes either; 57% a row within one lane. Those
377 lanes are a median 6.3 box heights tall (quartiles 3.7 and 9.4; 3% under two box heights), so
378 the row placement inside them was chosen, not forced. At most the choice of row count is
379 the lane's; where the box sits on that row, and the 43% held across or against the lanes, is
380 the person's.

381 6.0.0.7 The paired tests.

382 Per corpus, hand against `dot` on each criterion alone: the Hodges–Lehmann shift of $q_{0.02}$
383 with its 95% interval, zeros dropped, and the Holm-adjusted signed-rank p within the
384 corpus's family (`analyse.py`; the same families against `neato`, `prism` and `ELK` are in
385 `results/tests/`); the table is Appendix F.

386 The shifts under crossings, overlap, length and stress are distinguishable but confined to
387 the minority of diagrams where the pair differs at all. The large shifts are the profile's gaps:
388 the hand layout sits below `dot` on alignment in the biological corpora (-0.46 and -0.70) and
389 within $[-0.01, +0.09]$ of it on BPMN, a bound read off the interval, not an equivalence read
390 off the p of 0.14 (the random 300 below reads -0.05); below it on flow everywhere (-0.17 to
391 -0.40), and above it on node–edge separation and, in BPMN, on orthogonality ($+0.43$), the
392 criteria `dot` trades away.

393 6.0.0.8 The fourth corpus, restored.

394 The pre-registration named HOLA's formative drawings [13] and the previous draft dropped
395 them without a word. Restored: 136 drawings, 17 participants each laying out the same 8 ab-
396 stract graphs, scraped from the study's published SVGs (provenance beside `data/hola.txt`),
397 chord edges, below this paper's 15-box band and free of every diagram notation. The pattern
398 holds: overlap and crossings 1.00, length and stress 0.00, alignment 0.50, between Reactome's

0.21 and BPMN's 0.91. Node-edge holds 1.00, so H4 fails here too. Clustering and corpus agreement: Appendix J. These are drawings from a study tool rather than a diagram editor, which bears on the guide question of Section 7; the corpus does not record whether that tool drew alignment aids.

6.0.0.9 The motivating corpus.

The GD Collection itself [12] runs through the same test on the terms that need neither box extent nor edge direction: 1,072 of its 4,890 drawings have a largest component in the 15 to 40 band (chord edges, zero box sizes; provenance is not per drawing, and 69% of harvested figures reproduce their source in the collection's own audit; `results/gdcoll.py`). The profile is the paper's: length and stress hold 0.00, at exactly zero in 82 and 84% of drawings; crossings 1.00; orthogonality 0.06; alignment A1 holds 0.87 of the median drawing, with 0.43 of drawings at exactly 1.00 and 0.06 at zero. The corpus the standing proposal would fit first is held by the term the standard cost omits, at BPMN's level.

6.0.0.10 The 41 to 100 band.

The registered larger band, run against `neato` (`results/band41.md`; 108, 265 and 300 diagrams, the BPMN file the first of its band by id): every verdict stands. Length and stress hold 0.00 by hand in every corpus and `neato`'s stress holds 1.00; overlap 1.00; alignment 0.39, 0.16 and 0.74 against `neato`'s 0.07 to 0.08. The hold sits below the small band's 0.52, 0.21 and 0.91, so larger diagrams carry less alignment, and crossings stop holding every biological box (0.97 and 0.98): size buys improving crossing moves.

6.0.0.11 Null layouts.

Uniform random layouts of the same graphs put the chance line at 0.06 to 0.07 under alignment against hand 0.52, 0.21 and 0.91; length and stress hold 0.00 in the random layouts too, so the distance result separates optimisers from everyone, not humans from tools. A jitter of half a grid cell leaves alignment 0.76, 0.44 and 0.19 of its hold while orthogonality collapses, so orthogonality, not alignment, is the tie artefact. Axis-only moves still improve the distance terms, and the constrained model, stress subject to alignment [39, 40, 41], fails the same way: 0.87 to 0.90 of the median diagram improves without raising A1 (Appendix G).

6.0.0.12 The pre-registration's scorecard.

Four of the five registered hypotheses hold; H4, that alignment out-holds orthogonality and node-edge, is **refuted**, node-edge holding 0.88 to 1.00, and the omitted-term conclusion survives it because node-edge is priced in the standard energy and alignment is not (Appendix I). Registered and still not run: the caps in pitch units.

7 What else could produce this

1. **The editor snapped to a grid**, so alignment is the tool's. Excluded. The declared half-pitch displacement cannot fail at these pitches, for a corner reason Appendix G derives; the control that does discriminate is a layout snapped to the same grid, same boxes, same bounding box and no author, and it holds 0.065, 0.075 and 0.067 against the hand 0.52, 0.21 and 0.91. Displacing the hand layouts past $d/2$ takes them to that level,

XX:12 What Holds a Hand-Drawn Diagram?

and a random layout snapped to the plausible working pitch of five units holds 0.076, the same as at pitch one.

2. **The editor snapped to the neighbours**, so alignment is the guide's. Not separable on these corpora, and conceded. Snap-to-object guides escape any grid control, and the hold orders exactly as their affordances do: 0.87 to 0.91 where guides meet every drag, 0.52 where they are offered, 0.21 where they are absent. For fitting an energy the attribution changes nothing, the coordinates hold a term the energy lacks either way; for the word "optimise" it does, and the claim here is about the layouts, not the muscle. Separating them needs a guide-free editor or an interaction log, and these corpora carry neither. The nearest evidence is the restored HOLA corpus: drawings from a study tool, not a commercial editor, hold alignment at 0.50, so the phenomenon does not need Signavio to appear. The eras say the same (`results/era.md`): the GD Collection holds A1 at 0.81 in 1998 to 2005, when its figures came from xfig, Ipe and hand-coded PostScript with no shared guide, 0.60 in 2006 to 2014 and 0.95 in the TikZ years, where a typed coordinate is alignment chosen by hand; and WikiPathways is flat at 0.47 to 0.53 across four generations of WP numbers, so no gradient follows PathVisio's releases. No single editor's aid produces the hold; what stays open is only whether some aid does in each era.

The rest are excluded one by one in Appendix H, each with the test that did it and the numbers it turned on.

8 Related work

Aesthetic criteria and their weighting go back to Tamassia, Di Battista and Batini [3] and were made an explicit energy by Davidson and Harel [2]. Purchase [4] gave the criteria their metrics, and that list has no alignment metric. Whether people agree with the criteria has been asked three ways.

Three lines of evidence exist, tabulated with what each gives and leaves in Appendix L. By having people draw: van Ham and Rogowitz [16], Dwyer et al. [17], Purchase, Pilcher and Plimmer [25], HOLA's formative study [13], Mooney et al. [11], and the harvested collection [12]. By showing people drawings: [18, 14]. By learning from examples: [15, 26, 19, 28, 42] and O'Donovan, Agarwala and Hertzmann [8], who fit a design energy taking each example as optimal: the assumption examined here, and the closest prior instance of it we found.

Alignment as a human criterion is therefore known since 2012 in graph drawing, and since 1985 in drawing editors: Pavlidis and Van Wyk's beautifier inferred alignment constraints from the near-alignments of a rough drawing [37], Igarashi et al. made the inference interactive [38], and the guides of Section 7 are their descendants. The new part is the count of how many hand-placed boxes it holds, on diagrams of 15 to 40 boxes, beside the counts for the criteria tools weight. Two results bound what such a count can mean: van Wageningen, Mchedlidze and Telea [20] show drawings with identical metric values that differ in quality, so a fitted energy with no residual would still not be the energy people use; the claims here stay on the negative side; and Kobourov, Pupyrev and Saket [27] found crossings matter little on sparse large graphs, consistent with the sparsity result.

The machinery is standard elsewhere. Inverse optimisation, the recovery of an objective from solutions taken as optimal, is due to Ahuja and Orlin [22] and Keshavarz, Wang and Boyd [23]; Chan, Lee and Terekhov [24] define the residual reported here. Matched-budget random controls are standard in hyperparameter search [6] and were prescribed as the baseline for genetic algorithms in 1995 [7]; we have not found them in layout. No paper

fitting energy weights to human-placed coordinates was found in four searches of 2024 to 2026 work.

9 Conclusion

Coordinates a person accepted are not minima of the energy in use, and the way they fail is specific: exact under overlap, nowhere near under uniform edge length or stress, and held instead by an alignment into rows and columns that, HOLA's gridiness step aside [13], no standard energy prices.

A method that learns weights from such layouts must offer alignment among its terms or it fits nothing the person did. For a tool builder that is one addition and one number, not a subtraction: edge length and stress keep their weight, since a term inert at human coordinates is not thereby unwanted and the preference evidence runs the other way [11, 17]. The number to report is the fraction of boxes the energy holds when started from a layout a person accepted: $q_{0.02}$ as Section 5 fixes it, sixteen directions, fitted L , alignment A1, on layouts held out from any fit, with the decrease and active columns beside it, since the degenerate fit of Section 6 reaches 0.99 by weighting a term everyone satisfies and a bare held fraction can always be bought that way. The charge is also bounded: dot and ELK hold alignment at 0.42 to 1.00 without pricing it, layering aligns as a side effect, so the missing term convicts the force-directed and multicriteria energies, not the layered engines.

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XX:14 What Holds a Hand-Drawn Diagram?

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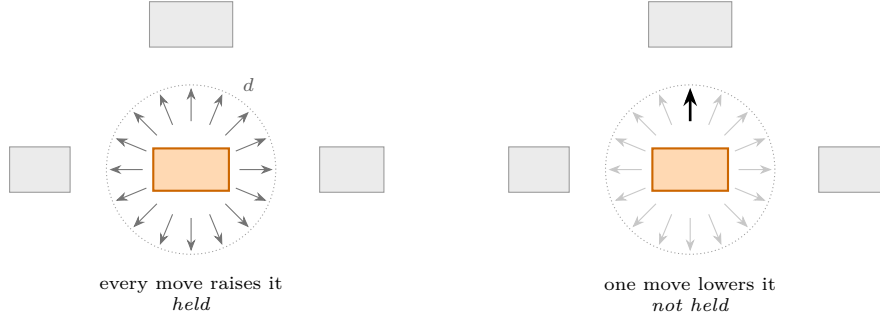


Figure 1 The test, one box and one criterion. The box is moved a distance d , two per cent of the drawing's width, in each of sixteen exact directions, and the criterion is recomputed. A box is held when no move lowers it.

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A The terms in full

term	definition	zero when
T_C crossings	proper crossings between segments of two routes with no shared endpoint, $/m$	planar
T_O overlap	overlap area of boxes / total box area	no overlap
T_L length uniformity	mean over edges of $((\ell - L)/L)^2$	all edges of length L
T_R orthogonality	mean over edges of the length-weighted mean over its segments of $\min(\Delta x , \Delta y)/(\Delta x + \Delta y)$	routes axis-aligned
T_A alignment	1 – fraction of nodes in an alignment of ≥ 3 (tol. 0.005)	every node aligned
T_N node–edge	squared shortfall of node to foreign route distance below $(w + h)/4$	none within
T_F flow	mean over directed edges of the backward component along the reading direction u , over length	no edge runs back

B The sums under composite energies

At the hand layout the three base terms share the energy as length 0.99, crossings 0.01, overlap 0.00, and $C + O + L$ inherits the length term's zero.

XX:16 What Holds a Hand-Drawn Diagram?

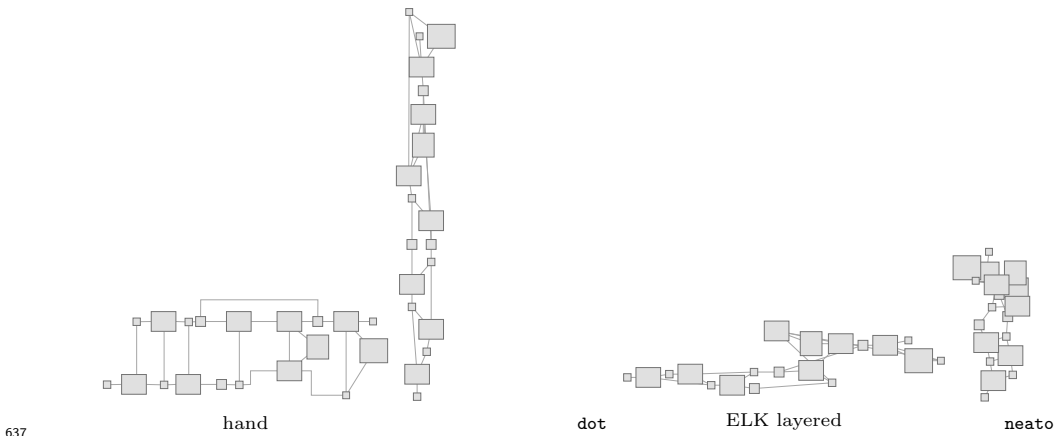


Figure 2 The same graph, the same box sizes, four layouts, one scale. The hand drawing and the two layered tools share the profile the test reads: held under overlap, free under length and stress. **neato** minimises stress and is the control that gives the test its power, held at 1.00 under stress where every hand layout reads 0.00. The tool layouts carry no waypoints, so their edges are drawn as chords.

609

	corpus	energy		hand	neato	prism	dot	elk
610	WikiPathways	C+O	1.00	[1.00, 1.00]	0.60	1.00	1.00	1.00
611		C+O+L	0.00	[0.00, 0.00]	0.17	0.19	0.00	0.00
612		C+O+S	0.00	[0.00, 0.00]	0.60	0.94	0.00	0.00
613		C+O+R	0.21	[0.19, 0.23]	0.00	0.00	0.19	0.22
614		C+O+A1	0.51	[0.47, 0.53]	0.04	0.07	1.00	0.57
615		C+O+A3	0.24	[0.22, 0.27]	0.03	0.03	0.50	0.41
616		C+O+grid	0.84	[0.81, 0.87]	0.45	0.79	0.96	0.74
617		C+O+N	0.86	[0.83, 0.88]	0.55	1.00	0.78	0.74
618		C+O+F	0.56	[0.53, 0.59]	0.30	0.50	1.00	1.00
619	Reactome	C+O	1.00	[1.00, 1.00]	0.76	1.00	1.00	1.00
620		C+O+L	0.00	[0.00, 0.00]	0.33	0.27	0.00	0.00
621		C+O+S	0.00	[0.00, 0.00]	0.76	0.89	0.00	0.00
622		C+O+R	0.17	[0.16, 0.19]	0.00	0.00	0.14	0.13
623		C+O+A1	0.22	[0.19, 0.25]	0.05	0.07	1.00	0.43
624		C+O+A3	0.11	[0.09, 0.12]	0.03	0.03	0.42	0.33
625		C+O+grid	0.64	[0.61, 0.67]	0.52	0.75	0.94	0.73
626		C+O+N	0.94	[0.91, 0.95]	0.71	1.00	0.80	0.75
627		C+O+F	0.61	[0.58, 0.65]	0.33	0.44	1.00	1.00
628	BPMN	C+O	1.00	[1.00, 1.00]	0.41	1.00	1.00	1.00
629		C+O+L	0.00	[0.00, 0.00]	0.24	0.29	0.00	0.00
630		C+O+S	0.00	[0.00, 0.00]	0.43	0.92	0.00	0.00
631		C+O+R	0.73	[0.71, 0.75]	0.00	0.00	0.28	0.40
632		C+O+A1	0.89	[0.86, 0.95]	0.04	0.09	0.88	0.68
633		C+O+A3	0.33	[0.30, 0.37]	0.03	0.03	0.33	0.45
634		C+O+grid	0.95	[0.93, 0.96]	0.28	0.78	0.75	0.67
635		C+O+N	1.00	[1.00, 1.00]	0.40	1.00	0.82	0.76
636		C+O+F	0.84	[0.81, 0.87]	0.26	0.61	1.00	1.00

C Decrease and active, term by term

The gridiness value puts 58%, 36% and 82% of hand-placed boxes in an alignment of three or more, none of **neato**'s, 88%, 82% and 78% of **dot**'s and 80%, 69% and 75% of **ELK**'s. The gridiness q of 0.77 to 0.78 on **neato**'s layouts, at value 1.000, is the step term's flatness, so alignment is compared on the value. Node-edge separation is nearly satisfied in every layout

(0.000 to 0.021) and holds 0.74 to 1.00 for the same reason. Crossings hold every box at this radius; the median value on the routes is 0.000 by hand in every corpus and 0.000 for `neato`, 0.029, 0.056 and 0.000 for `dot`, 0.036, 0.066 and 0.000 for ELK. Under flow `dot` and ELK are held at 1.00 with the term at 0.000 in every corpus; the hand layouts hold 0.59, 0.62 and 0.85 at 0.131, 0.096 and 0.045, and `neato`, which knows no direction, 0.44 to 0.62 at 0.121 to 0.222.

C.0.0.1 Step terms.

A count (crossings) or a step function of position (gridiness, Section 5) is flat almost everywhere and holds a node wherever no move of length d changes it. There q_d measures how much of the layout the criterion has stopped acting on, not how well it is met, so the term's value is reported beside q_d .

An interval of [0.00, 0.00] or [1.00, 1.00] is a pileup at the boundary, not precision: 87 to 92% of hand diagrams sit at exactly 0.00 under length and stress, and 62 to 85% of `neato`'s at exactly 1.00 under stress (`results/robustness.md`).

C.0.0.2 What q counts, term by term.

A held fraction is not commensurable across terms of different shape, and reading down a column compares three different things. Beside each q the table gives *dec*, the median over diagrams of the per-diagram mean best decrease a single move finds as a fraction of the term's value, and *active*, the share of diagrams in which any box has an improving move at all. Where *active* is small the term is flat at this radius: it holds every box without being satisfied by any of them, and its 1.00 carries no information about the person.

	WikiPathways			Reactome			BPMN		
term	q	dec	active	q	dec	active	q	dec	active
crossings	1.00	0.000	30%	1.00	0.000	20%	1.00	0.000	3%
overlap	1.00	0.000	18%	1.00	0.000	4%	1.00	0.000	7%
node-edge	0.88	0.047	64%	1.00	0.000	49%	1.00	0.000	15%
length	0.00	0.013	100%	0.00	0.013	100%	0.00	0.012	100%
stress	0.00	0.005	100%	0.00	0.006	100%	0.00	0.007	100%
orthogonality	0.19	0.012	100%	0.17	0.027	100%	0.72	0.008	99%
alignment A1	0.52	0.066	92%	0.21	0.069	99%	0.91	0.032	56%
flow	0.59	0.010	90%	0.62	0.012	95%	0.85	0.005	74%

D Sensitivities

The attachment point matters: of the 2,584 BPMN routes that carry an interior waypoint, 0.27 are axis-aligned at every segment when the route is closed at the box centres and 0.88 when it is closed at the stored attachments.

This is not a new device. Minimising a stress term over a scale factor is Smelser, Miller and Kobourov's scale-normalised stress [9], who give the minimiser in closed form and read it as a correlation coefficient, and Ahmed et al. [10] derive the same for the δ_{ij} -weighted graph stress used here. Their result is also why the fixed- L reading above is not a curiosity: under scale-sensitive stress they find a random layout ranked ahead of both `neato` and SFDP on nearly every graph of two 1,000-graph benchmarks. A control that reads 0.07 at a fixed L is that pathology, and fitting L is the known repair.

XX:18 What Holds a Hand-Drawn Diagram?

D.0.0.1 Radius.

q_d at $d \in \{0.005, 0.01, 0.02, 0.05\}$, hand layouts, then **neato**'s (`profile.py -sweep`):

	WikiPathways				Reactome				BPMN			
	.005	.01	.02	.05	.005	.01	.02	.05	.005	.01	.02	.05
energy												
overlap, hand	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
length, hand	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
stress, hand	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
alignment A1, hand	0.52	0.54	0.52	0.44	0.12	0.17	0.21	0.19	0.90	0.91	0.91	0.87
gridiness, hand	0.94	0.91	0.87	0.75	0.82	0.75	0.65	0.57	1.00	1.00	0.95	0.88
flow, hand	0.59	0.59	0.59	0.59	0.62	0.62	0.62	0.63	0.85	0.85	0.85	0.85
overlap, neato	0.60	0.60	0.60	0.60	0.75	0.76	0.76	0.76	0.41	0.41	0.41	0.43
length, neato	0.04	0.12	0.24	0.47	0.07	0.19	0.35	0.58	0.08	0.22	0.42	0.72
stress, neato	0.90	0.96	1.00	1.00	0.96	1.00	1.00	1.00	0.94	0.97	1.00	1.00
alignment A1, neato	0.04	0.05	0.06	0.06	0.03	0.05	0.06	0.07	0.03	0.05	0.06	0.07
gridiness, neato	0.93	0.86	0.78	0.65	0.90	0.83	0.77	0.61	0.92	0.85	0.78	0.67
flow, neato	0.50	0.50	0.50	0.50	0.45	0.45	0.45	0.45	0.62	0.62	0.62	0.62

The hand verdicts do not move with the radius. **neato**'s under length do: a smooth term near a minimum with gradient g and curvature λ holds a node when $|g| < d\lambda/2$, so a layout that is nearly a minimum is held at a large radius and free at a small one, and **neato**'s layouts, which minimise stress and not length uniformity, read as nearly uniform at $d = 0.05$ (0.47 to 0.70) and not at $d = 0.005$ (0.04 to 0.08). A hand layout at 0.00 across the range is far from the minimum at every scale the test resolves. Gridiness falls with d in every layout because a larger move reaches an alignment from further away; flow, a projection, does not move with d at all. ELK's block (`results/sweep.md`) reads as dot's: length and stress 0.00 at every radius, flow 1.00 at every radius.

D.0.0.2 Directions.

Each doubling of V contains the previous set, so q can only fall as directions are added and the primary sixteen-direction figures are upper bounds on the continuum value (`results/robustness.md`). At 8, 16, 32 and 64 directions the verdicts stand: hand length and stress 0.00 in every corpus at every count, **neato**'s stress 1.00 at every count. The alignment level is what moves: A1 by hand reads 0.58, 0.52, 0.47, 0.39 on WikiPathways, 0.31, 0.21, 0.14, 0.07 on Reactome and 0.92, 0.91, 0.90, 0.86 on BPMN across the four counts, while **neato**'s 0.06 falls to 0.00. The hand-to-control gap survives at 64; the levels are comparable only at one count and are stated at sixteen. The full grid of radius, direction count, reference-length rule and route treatment, 1,008 cells, flips no verdict: under the fitted L the hand distance terms read exactly 0.00 in every cell (`results/specgrid.md`). The tie tolerance, the last instrument constant, moves nothing at all: the A1 medians are identical at 0.001 through 0.01 (`results/tolsweep.md`), so the alignments the test counts are exact ties, not tolerance-window artefacts.

D.0.0.3 Reference length.

Under the median edge length instead of the fitted L , **neato**'s stress layouts score 0.07, 0.08 and 0.12 under stress and the hand layouts 0.00; under length the hand layouts score 0.03, 0.04 and 0.05 and **neato**'s 0.26, 0.41 and 0.45. Under $L = 1/\sqrt{N}$ no box of any layout is held under either term, the scale being wrong for every layout. The hand verdicts do not depend on the choice; the control's does, so the fit is primary (Section 2).

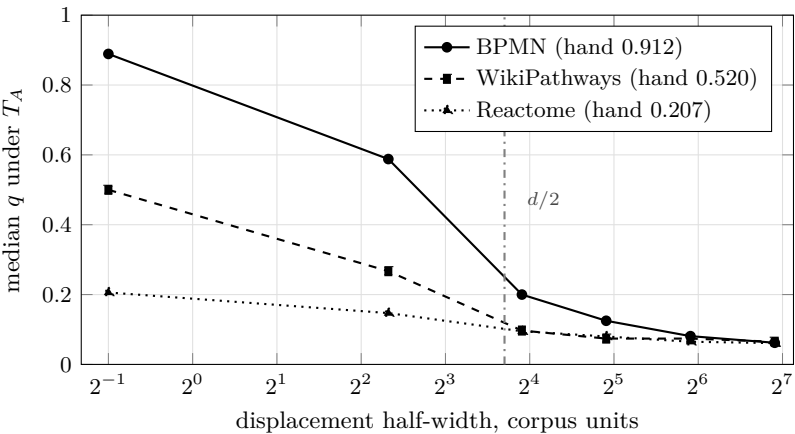


Figure 3 The jitter sweep. The declared half-pitch test sits at the left edge, 0.5 units, where nothing has changed. The corner of T_A protects a single displaced box below $\epsilon = d/2$, but every box is displaced, relative offsets reach 2ϵ , and the hold falls well before $d/2$: BPMN 0.89 to 0.59 by a half-width of 5. By $d/2$ every corpus is at the grid-snapped level of 0.06 to 0.08.

D.0.0.4 Box extent.

Centre distances on box diagrams could be the objection: with large, uneven boxes a person could sit at a minimum of a border-to-border distance term while failing the centre one. Measured border to border, each box’s offset to its border along the joining direction subtracted and L refitted on those distances, the hand verdicts do not move: length and stress hold 0.00 in every corpus, 0.86 to 0.92 of diagrams at exactly zero (`results/border.py`). `neato`, which minimised the centre form, is not held under the border form (0.00 to 0.09), as a layout optimising one criterion and tested on another should not be.

D.0.0.5 Routes.

The tables of Section 6 are on the stored routes; the chords are the sensitivity, `chords.md` beside the tables in the repository. Closing every edge at the box centres moves no verdict: crossings still hold every box (the WikiPathways value rises from 0.000 to 0.038, one crossing per 26 edges, the anchors’ work), orthogonality falls from 0.19, 0.17 and 0.73 to 0.16, 0.05 and 0.40, and node–edge separation from 0.88, 1.00 and 1.00 to 0.76, 0.85 and 0.87. The drawn geometry only raises the hand layout’s standing on the criteria it touches.

Each format’s stored routes, as a fraction of the edges in the band:

corpus	edges	with a waypoint	by what the file stores
WikiPathways	8,253	0.291	chord 0.587, anchor 0.225, elbow 0.154, curved 0.026, segmented 0.008
Reactome	6,827	0.069	chord 0.931, stored 0.069
BPMN	7,910	0.327	chord 0.673, stored 0.327

GPML’s Elbow and Curved connectors store their endpoints only and PathVisio computes the bends at draw time, so those routes read as chords; a quarter of WikiPathways edges end on an anchor of another interaction, where the person drew no segment to the node and the last segment is manufactured. Only the BPMN routes are orthogonal: of the WikiPathways and Reactome routes with a waypoint, 0.009 and 0.007 are axis-aligned at every segment, the biological formats drawing diagonal segments. The chords are the sensitivity.

765 **E** Descent, chance and the displacement yardstick

766 **E.0.0.1 The descent, in full.**

767 Through the library: hill climbing from the hand layout, one node per proposal, first step
 768 $d/2$ per coordinate, the step halved after 200 rejected proposals, each node kept within a disc
 769 of radius d about its start by the repair, 4000 energy evaluations, 15 seeds per instance; and
 770 uniform sampling in the same cap at the same budget on the same seeds, the matched-budget
 771 control. Reported per instance: the fraction of the cap used (mean over nodes of the distance
 772 moved over d), the fraction of energy removed, for the climber and for the control, and the
 773 number of seeds on which the climber ends lower than the control with its exact sign test.
 774 The climber pilot used a step of 0.1, five times the cap, so that nearly every proposal landed
 775 on the cap circle; the base-energy result did not depend on it (98.5 to 95.9% of the cap used
 776 as the step fell from 0.1 to 0.005) and the alignment result did (47 to 18%), so the directional
 777 test is primary. The converged reference is the same climber uncapped, 60,000 evaluations,
 778 with q_d measured where it stops.

779 **E.0.0.2 Descent and chance.**

780 The secondary estimand, per corpus and energy: the median fraction of the cap the capped
 781 climber uses and of the energy it removes, the same for uniform sampling in the cap at the
 782 same budget and seeds, the share of diagrams where the climber ends lower on at least 12 of
 783 15 seeds, and $q_{0.02}$ where the uncapped climber stops, the converged reference.

784 corpus	785 energy	cap used		energy removed		$\geq 12/15$	q conv.
		climb	random	climb	random		
786 WikiPathways	C+O+L	0.99	0.76	0.28	0.16	0.88	1.00
	C+O+A1	0.39	0.74	0.71	0.17	0.75	1.00
	stress	1.00	0.77	0.12	0.05	1.00	1.00
	C+O+F	0.44	0.76	0.27	0.18	0.77	1.00
790 Reactome	C+O+L	1.00	0.76	0.31	0.16	0.93	1.00
	C+O+A1	0.56	0.74	0.99	0.37	0.80	1.00
	stress	1.00	0.77	0.13	0.06	1.00	1.00
	C+O+F	0.36	0.75	0.32	0.22	0.71	1.00
794 BPMN	C+O+L	1.00	0.77	0.26	0.12	0.99	1.00
	C+O+A1	0.11	0.73	0.24	-0.01	0.95	1.00
	stress	1.00	0.77	0.14	0.07	1.00	1.00
	C+O+F	0.16	0.74	0.09	0.04	0.66	1.00

798 Under the distance energies the climber walks every node to the cap circle (0.99 to 1.00
 799 of d) and is still removing energy when the budget ends; under C+O+A1 it moves 0.11 to
 800 0.56 of the cap, the misalignments being nearer than d . The climber beats the control on all
 801 15 seeds, which survives Holm over the corpus (exact sign $p = 3.1 \times 10^{-5}$ against 0.05/305),
 802 on 0.81 to 0.99 of diagrams under C+O+L and on all of them under stress. The converged
 803 reference's median q is 1.00 for every energy and corpus, the test's power on the sums; the
 804 fraction of diagrams whose reference stops below 0.9 is at most 0.02 everywhere but C+O+F
 805 (0.05, 0.07 and 0.13), the flow term's plateaus.

806 **E.0.0.3 The displacement yardstick.**

807 Perturb `neato`'s converged stress layouts, every box moved a known fraction of the width in
 808 a seeded direction, and run the same capped climber (`results/calib.py`). Unperturbed it
 809 removes 0.001 of the stress energy; at a displacement of 0.005, 0.007 to 0.024; at 0.02, 0.09

to 0.26; at 0.05, 0.24 to 0.43; at 0.10 less again, the cap seeing a shrinking share of a growing energy. The hand layouts' 0.12 to 0.14 under stress sits at a displacement of 0.014 to 0.025.

F The paired tests in full

F.0.0.1 The statistics, in full.

The unit of inference is the diagram. Per corpus, the paired comparison of q_d between the hand layout and a tool control is the Wilcoxon signed-rank test, zeros dropped, ties by midranks, exact on the observed ranks up to 50 non-zero pairs and the normal approximation with continuity and tie corrections above, with the exact sign test beside it and the Hodges–Lehmann shift with its distribution-free 95% interval; the family is the terms compared against one control within one corpus, Holm-corrected. The script carries 630 self-checks against enumeration and against R. Equivalence claims (“no reweighting repairs it”) are read off the sweep, never off a failure to reject.

The columns are those of Section 6: the Hodges–Lehmann shift of $q_{0.02}$, hand against dot, with the Holm-adjusted signed-rank p .

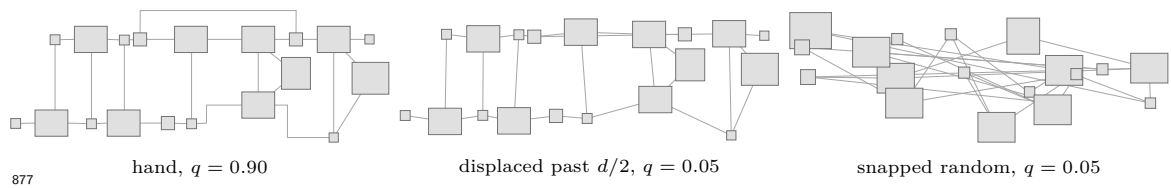
824 energy	WikiPathways				Reactome				BPMN			
	HL [95%]		p		HL [95%]		p		HL [95%]		p	
crossings	−0.04	[−0.07, −0.01]	0.016	+0.02	[−0.01, +0.05]	0.30	+0.10	[+0.08, +0.12]	<1e-4			
overlap	−0.14	[−0.18, −0.11]	<1e-4	−0.07	[−0.10, −0.06]	0.0024	−0.22	[−0.29, −0.11]	<1e-4			
length	−0.01	[−0.03, +0.00]	0.27	−0.03	[−0.05, −0.01]	0.017	−0.09	[−0.12, −0.06]	<1e-4			
stress	−0.03	[−0.04, −0.00]	0.022	−0.03	[−0.05, −0.01]	0.0048	−0.08	[−0.11, −0.05]	<1e-4			
orthogonality	+0.01	[−0.00, +0.04]	0.27	+0.02	[+0.00, +0.04]	0.036	+0.43	[+0.41, +0.45]	<1e-4			
alignment A1	−0.46	[−0.49, −0.43]	<1e-4	−0.70	[−0.73, −0.67]	<1e-4	+0.04	[−0.01, +0.09]	0.14			
alignment A3	−0.25	[−0.28, −0.22]	<1e-4	−0.28	[−0.31, −0.26]	<1e-4	+0.13	[+0.10, +0.16]	<1e-4			
gridiness	−0.13	[−0.15, −0.11]	<1e-4	−0.29	[−0.32, −0.26]	<1e-4	+0.21	[+0.17, +0.25]	<1e-4			
node-edge	+0.13	[+0.10, +0.16]	<1e-4	+0.18	[+0.15, +0.21]	<1e-4	+0.26	[+0.23, +0.28]	<1e-4			
flows	−0.40	[−0.42, −0.38]	<1e-4	−0.33	[−0.35, −0.31]	<1e-4	−0.17	[−0.19, −0.14]	<1e-4			

G Null layouts in full

G.0.0.1 The grid snap exclusion, in full.

Excluded by the control the pre-registration named and by a stronger one it did not. The detected pitch is one unit in Reactome and BPMN (67% and 63% integer coordinates) and none in WikiPathways (17%, against 0.37 of boxes exactly aligned in x). Displacing every box by up to half a pitch leaves the held fraction at 0.500, 0.206 and 0.889 against 0.520, 0.207 and 0.912. That test cannot fail at this pitch, and the reason is the instrument: T_A is $\min_j \min(|\Delta x|, |\Delta y|)$, so a box offset by ϵ from its nearest agreement is released only when a move of radius d lowers the term, and moving d toward the agreement lands at $d - \epsilon$, which is worse whenever $\epsilon < d/2$. Half a pitch is 0.5 units against a $d/2$ of 8, 8 and 13. The control that does discriminate is a layout snapped to the same grid, with the same boxes and the same bounding box and no author: it holds 0.065, 0.075 and 0.067 (Figure 4). Displacing the hand layouts past $d/2$ takes them to that same level (0.097, 0.095 and 0.200 at a half-width of 15, 0.065, 0.061 and 0.062 at 120). The grid is available to the random layout and does not produce the effect. Two additions after review: a random layout snapped to the plausible working pitch of five units holds 0.076, the same 0.07 as at pitch one, so the null is not flattered by the finest pitch; and the jittered hand layouts begin falling well below $d/2$ (0.59 on BPMN at a half-width of 5), so the corner argument bounds only the declared half-pitch test, and the caption of Figure 3 says so.

XX:22 What Holds a Hand-Drawn Diagram?



878 **Figure 4** The grid alternative of Section 7: the grid alone does not produce the alignment hold.
879 One BPMN model, 19 boxes; the corpus medians are 0.91, 0.20 and 0.07. Left, the drawing as its
880 author left it. Centre, every box displaced by up to 15 units, just past the $d/2$ below which the
881 directional test cannot resolve a misalignment. Right, every box placed uniformly at random in the
882 same bounding box and snapped to the same grid: the grid is available and the author is not. Box
883 sizes are unchanged throughout. The hand panel draws the routes the file stores; the two control
884 panels draw chords, their boxes no longer being where the routes ran.

855 G.0.0.2 Null layouts.

856 Review asked for the chance line under every term, not only alignment. Uniform random
857 layouts of the same graphs, chord geometry, one draw per diagram, seeds from the diagram
858 ids: alignment 0.06 to 0.07 against hand 0.52, 0.21 and 0.91; node-edge 0.11 to 0.12 against
859 0.76 to 0.87; flow 0.26 to 0.41 against 0.59 to 0.85; orthogonality 0.00 against 0.05 to 0.40.
860 Length and stress hold 0.00 in the random layouts too: no layout of any provenance is
861 stationary under a distance term at this radius except a tool's converged output, so the
862 distance result separates optimisers from everyone, not humans from tools. Crossings and
863 overlap hold 0.58 to 0.67 of random boxes, the flat base rate the active column already prices.
864 Under a jitter of half a grid cell either way the alignment hold keeps 0.76, 0.44 and 0.19 of its
865 three corpora while orthogonality falls from 0.40 to 0.18 and 0.16 to 0.06: orthogonality, not
866 alignment, is the tie artefact. Axis-only moves, which preserve every perpendicular alignment
867 tie, still improve length and stress on the median diagram of every corpus (q 0.00, means
868 0.005 to 0.016), and a weight of 0.001 on length collapses the held fraction of $C+O+A_1$ from
869 0.89, 0.51 and 0.22 to 0.13, 0.11 and 0.07: the distance terms are unpriced even inside the
870 alignment-preserving move set, and no small weight on them survives beside alignment. The
871 constrained model of the layout literature, stress minimised subject to alignment constraints
872 [39, 40, 41], fails under its own move set: restricted to the directions that do not raise A_1 ,
873 0.87 to 0.90 of the median diagram's boxes still have a stress-lowering move, and the boxes
874 alignment pins entirely, no direction admissible, are 0.02 to 0.05 (`results/constrained.py`).
875 The generator and every number: `results/nulls.py`.

876 H The other alternatives

885 The rest, each with the test that excluded it and the numbers it turned on:

alternative	test	result
Reference length	L at the median edge length and at $1/\sqrt{N}$	hand held on 0.00 to 0.06 of boxes under length and stress, as under the fit; only the control moves
Straight segments where the person drew polylines	primary tables on the stored routes, chords beside them	no verdict moves
A constrained energy whose constraints are absent	the flow term in the tables; the BPMN lanes decomposed	flow holds 0.59 to 0.85 of hand-placed boxes, so reading direction is a criterion partly satisfied and not the missing constraint; 43% of held BPMN boxes are held by an alignment no horizontal lane can impose, and the lanes of the rest leave a median 6.3 box heights of room. Containment as a hard constraint was not added
The radius	$d \in \{0.005, 0.01, 0.02, 0.05\}$; at $d = 0.02$ the move is about one grid step	the hand verdicts are the same at every d . The table at d in units of box width and of median edge length was not run
Selection: near-trees make overlap and crossings easy	q against m/n over diagrams (<code>sparsity.py</code>)	crossings do track density (Spearman ρ $-0.37, -0.34, -0.11$ by hand and $-0.48, -0.30, -0.41$ for <code>dot</code>), these being near-trees at m/n 1.06, 1.04 and 1.11 against 1.30 in the Rome benchmark [21], so “crossings hold every box” is partly the trees’ gift, given to every layout of the same graphs alike; crossings are what the theory is about [1], and on these diagrams there are few to remove. Overlap does not ($+0.02, -0.06, +0.12$), and length and stress are 0.00 in the sparse and the dense half both. No hand-tool difference is explained
The tools are the wrong tools	ELK layered 0.12.0, the engine the BPMN and KIELER editors run, as a fourth control	its profile is <code>dot</code> ’s: length and stress 0.00, flow 1.00 at value 0.000, alignment 0.42 to 0.62 where <code>dot</code> reaches 0.89 to 1.00

I The pre-registration’s scorecard

An earlier version of this study took 147 BPMN models without the connector filter, two fifths of them read as bipartite graphs.

I.0.0.1 The pre-registration’s scorecard.

H1, crossings and overlap hold the hand layout above 0.9 and above `neato`: holds, 1.00 against 0.41 to 0.76. H2, length alone holds under 0.2 and below `neato`: holds, 0.00 against 0.24 to 0.42. H3, no positive length weight repairs $C+O+L$: holds, fitted weight 0.000 everywhere and q 0.008 at any weight past 0.001. H4, alignment holds more boxes than orthogonality or node-edge in every corpus: **refuted**. Node-edge holds 0.88, 1.00 and 1.00 of hand-placed boxes against alignment’s 0.52, 0.21 and 0.91. Node-edge is priced in the standard energy and alignment is not, so the omitted-term conclusion stands, but the registered ranking was wrong and the recorded prediction with it. H5, exploratory, stress holds the hand layout less than `neato`’s: holds, 0.00 against 1.00. The registered fourth corpus, HOLA’s formative drawings, was dropped undeclared from the previous draft; it is restored above. The deduplication pass is run in Section 4 and the 41–100 band above. Registered and still not run: the caps in pitch units.

XX:24 What Holds a Hand-Drawn Diagram?

909 J Deduplication and drawers

910 194 of the 248 Reactome graphs and 120 of the 300 BPMN graphs in the band are the
911 component of a larger diagram; no WikiPathways pathway in the band is.

912 The 136 drawings are 17 participants crossed with 8 graphs, so the interval is clustered:
913 resampling participants (2000 draws) puts the alignment median at [0.31, 0.57], and the
914 per-participant medians run 0.14 to 0.96. Across the four corpora the seven terms every
915 corpus measures rank in the same order, Kendall's $W = 0.99$.

916 The registered deduplication pass (`results/dedup.py`): by exact coordinate multiset,
917 one Reactome pair is a single drawing stored under two ids; by the structural key (n , m ,
918 degree sequence, box sizes), one further Reactome pair and one WikiPathways pair; the 300
919 BPMN models collide under neither key, so the feared resubmission duplicates are absent
920 from this sample. Removing every extra member moves no headline median. Clustering by
921 drawer: the 305 WikiPathways pathways carry 153 first authors, the most prolific 9% of
922 the corpus; the A1 median's interval clustered by first author is [0.48, 0.55] against [0.48,
923 0.54] over diagrams, and the twenty drawers of three or more pathways hold medians 0.19
924 to 0.85 (`results/drawers.py`). Reactome curators and BPMN authors are not recorded in
925 the archives as parsed.

926 K Reproducibility

927 Parsers, the energy, the test, the descent, the fitting scripts, the parsed corpora and
928 every table's script are in the repository under `example/diagrams`; the main tables re-
929 build by `results/run_all.sh`, and every sensitivity of the appendix ships its script be-
930 side its committed output. `results/manifest.txt` records commit, platform, tool ver-
931 sions, seeds and the SHA-256 of the archives and the corpora as measured. The tracked
932 `data/bpmn_population_a1.csv` rebuilds the abstract's 0.87 from a clone; the repository
933 carries derived coordinates and ids, never archive content, and the two biological corpora are
934 CC-licensed exports.

L Related work, tabulated

work	what it did	what it gives, and what it leaves
<i>By having people draw</i>		
van Ham, Rogowitz [16]	73 hand layouts against a force-directed algorithm, half started from its optimum	71 of 73 raise edge-length variance above the optimum; 60% fewer crossings
Dwyer et al. [17]	32 participants laid out one graph; 194 judges ranked the results against three automatic layouts	judges' choices track stress, without a test; its Table 2 has user layouts at 3 to 9 times the force-directed stress
Purchase, Pilcher, Plimmer [25]	participants drew from adjacency lists	crossing removal the strongest aesthetic, alignment to a grid important
Kieffer et al. [13]	17 participants' layouts of 8 small graphs, ranked by 66 judges	gridiness correlates with rank on all 8, the source of HOLA's alignment step; rank tracks stress on 4 of 8
Mooney et al. [11]	preference and shortest-path performance against stress, graphs to 50 nodes	people prefer low stress and perform better as it falls; why the conclusion adds a term and removes none
Mooney et al. [12]	4,890 drawings harvested automatically from 27 editions of these proceedings	measured in Section 6: length and stress 0.00, alignment 0.87. Provenance is not per drawing and edge directions are dropped, so flow and the box terms are not run
<i>By showing people drawings</i>		
Purchase et al. [18]	a Turing test on hand and algorithmic drawings	hand drawings distinguishable, and judged better
Klammler, Mchedlidze, Pak [14]	Nelder-Mead fit of a linear aesthetic combination discriminating good from worsened layouts	no human coordinates enter it; they call human-labelled data a critically important accomplishment still to come
<i>By learning from examples</i>		
Masui [15]	a weighted constraint combination from good and bad layouts, by genetic programming	neither tested whether the examples are minima of the learned energy
Spönemann et al. [26]	weights adjusted from user selections among candidates	as above
Flud [19]	annealing continued from crowd layouts of pathway diagrams, three networks	the closest prior descent from a hand layout; it raised an asserted five-term score and read that as success. The cap, the per-term test, the reference and the controls here are what make it a measurement of the score
Zhang et al. [28]	64,222 pairwise preference labels	the nearest recent work; it learns from preferences, not coordinates
Wang et al. [42]	a graph network drawing arbitrary graphs under a weighted sum of aesthetics	the weights are asserted or tuned, not learned from people; no human coordinates enter it
O'Donovan, Agarwala, Hertzmann [8]	a 122-parameter energy over single-page graphic designs fitted to a handful of human examples by nonlinear inverse optimisation	each example taken as optimal under the unknown parameters: the assumption examined here, and the closest prior instance of it we found